

SPANNING TREES OF $G \times K_2$ BY VERTICAL SUPPORT, PARTIAL DOUBLES, AND THE BIJECTIVE PROGRAM FOR THE HYPERCUBE TREE FORMULA

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ABSTRACT. For a connected graph G , we count the spanning trees of the prism $G \times K_2$ by their *vertical support* — the set S of vertices of G whose vertical edge is used: the count is $\frac{1}{2} c(G) 2^{|S|} F_G(S)$, where $c(G)$ is the number of spanning trees and $F_G(S)$ the number of S -rooted spanning forests of G . Iterating this identity over the n directions of the hypercube $Q_n = Q_{n-1} \times K_2$ telescopes, through Pascal’s rule, to Stanley’s product formula $c(Q_n) = 2^{2^n - n - 1} \prod_k k^{\binom{n}{k}}$, giving an elementary inductive derivation of that formula and, more importantly, reducing the problem of a fully bijective proof — open since [2, Example 5.6.10], and still open in bijective form after Bernardi’s combinatorial proofs [1] — to a single per-level bijective step. We then prove the *class-level law*: refining by the multiset U of level-projected horizontal edges, the trees in each class (S, U) number exactly $d(S, U) 2^{|S|-1}$, where $d(S, U)$ counts the splittings of U into a spanning tree and an S -rooted forest of G — and each splitting occurs among the trees of its class as an explicit *polarized* representative. This proves every counting statement in the program down to its finest stratum; what remains open, and is stated precisely, is a canonical straightening map from arbitrary trees to polarized ones, which we supply in the acyclic-support case. We also record an exactly verified spin-uniformity phenomenon (Bernardi’s independence holds exactly for rooted-anywhere trees and provably fails for root-fixed ones) that constrains the shape of any such map.

1. BACKGROUND

Stanley’s *Enumerative Combinatorics* vol. 2 derives

$$(1) \quad c(Q_n) = 2^{2^n - n - 1} \prod_{k=1}^n k^{\binom{n}{k}}$$

from the Laplacian eigenvalues of Q_n and remarks (1st ed., p. 62, closing Example 5.6.10) that “a direct combinatorial proof of this formula is not known.” Bernardi [1] gave two combinatorial proofs — one via an independence property of edge-direction “spins” in uniformly random rooted forests, one via killing involutions — and the second edition of [2] (2024, Notes to Ch. 5) updates the remark: a *bijective* proof is not known. This paper organizes a route to one.

Throughout, G is a connected graph on M vertices (for (1), $G = Q_{n-1}$ and $M = 2^{n-1}$), and $G \times K_2$ is the prism over G : two copies $G \times \{0\}$, $G \times \{1\}$ (*horizontal* edges) plus a *vertical* edge $\{(u, 0), (u, 1)\}$ for each $u \in V(G)$. For a spanning tree T of $G \times K_2$, let $S(T) \subseteq V(G)$ be its vertical support. For $S \subseteq V(G)$, let $F_G(S)$ denote the number of spanning forests of G with exactly $|S|$ components, each containing one vertex of S ; by the all-minors Matrix-Tree

theorem $F_G(S) = \det L_G[\bar{S}]$, the minor of the Laplacian with the rows and columns of S removed.

2. THE SUPPORT-PRODUCT THEOREM

Theorem 2.1. *For every nonempty $S \subseteq V(G)$, the number of spanning trees T of $G \times K_2$ with $S(T) = S$ is*

$$N_G(S) = \frac{1}{2} c(G) 2^{|S|} F_G(S).$$

Proof. Weight the vertical edge at u by an indeterminate z_u and every horizontal edge by 1, and let $t(z) = \sum_T \prod_{u \in S(T)} z_u$ be the weighted spanning-tree sum. With vertices ordered level by level, the weighted Laplacian is $\begin{pmatrix} L_G + Z & -Z \\ -Z & L_G + Z \end{pmatrix}$, $Z = \text{diag}(z_u)$, which the orthogonal level-swap basis $\frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix}$ block-diagonalizes as $L_G \oplus (L_G + 2Z)$. For a weighted graph on N vertices the product of the nonzero Laplacian eigenvalues is N times the tree sum, and the kernel lies in the L_G block; with $N = 2M$,

$$t(z) = \frac{1}{2M} \left(\prod_{\lambda \neq 0} \lambda(L_G) \right) \det(L_G + 2Z) = \frac{c(G)}{2} \det(L_G + 2Z).$$

Expanding by principal minors in the diagonal matrix $2Z$ and applying the all-minors Matrix-Tree theorem, $\det(L_G + 2Z) = \sum_R 2^{|R|} F_G(R) \prod_{u \in R} z_u$. Both sides are multilinear in the z_u ; compare coefficients of $\prod_{u \in S} z_u$. $\square$

Remark 2.2. The identity is elementary and product-graph Matrix-Tree computations of this kind appear in Martin–Reiner [3] and Bernardi [1, §4]; we make no novelty claim for Theorem 2.1 itself. Its role here is architectural: it is the exact statement whose bijective realization would prove (1) bijectively (Corollary 2.3 and Problem 5.1). We first found it as an experimental identity: it was confirmed class-by-class for Q_2, Q_3, Q_4 (3+15+255 supports; the Q_4 case over all 679 477 248 tree–root pairs) before the proof was written down.

Corollary 2.3. *Iterating Theorem 2.1 yields (1).*

Proof. Summing the theorem over all S (the $S = \emptyset$ term vanishes) and using $\sum_R z^{|R|} F_G(R) = \det(zI + L_G)$ at $z = 2$:

$$c(G \times K_2) = \frac{1}{2} c(G) \det(2I + L_G).$$

For $G = Q_{n-1}$, whose Laplacian eigenvalues are $2k$ with multiplicity $\binom{n-1}{k}$, $\det(2I + L) = \prod_{k=0}^{n-1} (2k+2)^{\binom{n-1}{k}}$. Writing $R_n = 2^n c(Q_n)$, the recursion reads $R_n = R_{n-1} \prod_{j=1}^n (2j)^{\binom{n-1}{j-1}}$, and Pascal’s rule gives $R_n = \prod_{j=1}^n (2j)^{\binom{n}{j}}$ by induction, which is (1). $\square$

Of course, evaluating $\det(2I + L_{Q_{n-1}})$ spectrally re-uses linear algebra; combinatorially, $\det(2I + L_G) = \sum_S 2^{|S|} F_G(S)$ counts *pairs* (an S -rooted spanning forest, a $\{0, 1\}$ -coloring of its roots), so the recursion has the set-level form

$$(2) \quad \begin{aligned} & \{\text{spanning trees of } G \times K_2\} \times \{0, 1\} \\ \longleftrightarrow & \{\text{spanning trees of } G\} \times \{\text{root-2-colored rooted spanning forests of } G\}, \end{aligned}$$

and a bijective proof of (1) needs exactly a constructive form of (2), level by level.

3. BIJECTIVE FRAGMENTS

3.1. The full-support slice.

Proposition 3.1. *A spanning tree of $G \times K_2$ containing every vertical edge is the same data as a spanning tree of G together with a level in $\{0, 1\}$ for each of its $M - 1$ edges: contract the vertical edges to project, and lift each base tree edge to its recorded level to invert. Hence $N_G(V(G)) = c(G) 2^{M-1}$, the $S = V(G)$ case of Theorem 2.1.*

3.2. Partial doubles.

Definition 3.2. For $S \subseteq V(G)$, the *partial double* $G^{(S)}$ is the graph obtained from G by splitting each $u \notin S$ into u_0, u_1 : an edge (u, v) of G becomes two parallel edges when $u, v \in S$; the two edges $(u, v_0), (u, v_1)$ when $u \in S, v \notin S$; and $(u_0, v_0), (u_1, v_1)$ when $u, v \notin S$.

Contracting the vertical edges of a tree with support S is a bijection onto the spanning trees of $G^{(S)}$, so Theorem 2.1 is equivalent to

$$(3) \quad c(G^{(S)}) = \frac{1}{2} c(G) 2^{|S|} F_G(S).$$

For $S = V(G)$ this is Proposition 3.1 (all edges doubled); for $S = \emptyset$ both sides vanish on connectivity grounds. Splitting one vertex at a time interpolates between $G^{(V)}$ and $G^{(S)}$ and is a natural inductive route to a bijection, though the ratio $F_G(S)/F_G(S \cup u)$ is not local, so the induction must carry more structure than a single count.

3.3. What the spins do and do not give. Bernardi's independence theorem [1, Thm. 1] concerns uniformly random rooted forests conditioned on their projection; transported to spanning trees it says the *spins* (for each $u \in S$, the level of the endpoint of the vertical edge at u nearer the root) should be free bits. We verified computationally: for the ensemble of (tree, root) pairs of Q_3 , conditioned on the direction and the support S , the joint distribution of the spins is *exactly* uniform on $\{0, 1\}^S$ (all 45 conditioned classes, zero discrepancy), and remains consistent with uniform on samples of 2×10^6 (Q_4) and 10^6 (Q_5) uniform trees. For trees with a *fixed* root the uniformity provably fails (e.g. $\chi^2 = 330.7$ on 65 dof already for Q_3). Thus the factor $2^{|S|}$ of Theorem 2.1 is realized by free bits only in the rooted-anywhere ensemble; any bijection must fix the root only after the bits are extracted. This matches the shape of (2), which is stated for a 2-fold cover of the tree set rather than for rooted trees.

4. THE CLASS-LEVEL LAW

Fix S and refine further. Deleting (rather than contracting) the rungs of a tree T' with support S leaves horizontal forests in the two levels; projecting both to G gives an edge multiset U with multiplicities in $\{1, 2\}$: write D for the doubled edges (used in both levels), M for the singletons ($m = |M|$, each carrying a free bit: which level), and $N(S, U)$ for the number of trees with data (S, U) . Call a pair (T, F) a *decomposition* of (S, U) if T is a spanning tree of G , F is an S -rooted spanning forest of G , and $T \uplus F = U$ as multisets; let $d(S, U)$ be their number. Note that a decomposition determines its class: $D = T \cap F$ and $M = T \triangle F$.

Theorem 4.1 (class-level law). *For every connected G , every nonempty $S \subseteq V(G)$, and every multiset U ,*

$$N(S, U) = d(S, U) \cdot 2^{|S|-1}.$$

Proof. Introduce a variable w_e for each edge e of G and z_u for each $u \in V(G)$; in $G \times K_2$ weight both copies of e by w_e and the rung at u by z_u . As in Theorem 2.1, the weighted Laplacian block-diagonalizes as $L_w \oplus (L_w + 2Z)$, where L_w is the weighted Laplacian of (G, w) and $Z = \text{diag}(z_u)$, and the weighted tree sum of the prism, $t(w, z) = \sum_{T'} \prod_e w_e^{\#\text{copies of } e \text{ in } T'} \prod_{\text{rungs } u \in T'} z_u$, equals $\frac{1}{2} \tau(w) \det(L_w + 2Z)$ with $\tau(w)$ the weighted spanning-tree sum of G . Expanding the determinant by principal minors in $2Z$ and applying the weighted all-minors Matrix-Tree theorem,

$$t(w, z) = \frac{1}{2} \tau(w) \sum_{R \subseteq V} 2^{|R|} \varphi_R(w) \prod_{u \in R} z_u,$$

where $\varphi_R(w)$ is the weighted R -rooted spanning-forest sum. Both sides are multilinear in the z_u ; the coefficient of $\prod_{u \in S} z_u$ gives

$$\sum_{T': S(T')=S} \prod_e w_e^{\#\text{copies}} = 2^{|S|-1} \tau(w) \varphi_S(w).$$

Now extract the coefficient of $w^U := \prod_{e \in D} w_e^2 \prod_{e \in M} w_e$. On the left this is $N(S, U)$. On the right, τ and φ_S are multilinear in the w_e (a tree or forest uses each edge at most once), so the coefficient of w^U in the product counts exactly the pairs (T, F) with $T \uplus F = U$, i.e. $d(S, U)$. $\square$

Setting $w \equiv 1$ recovers Theorem 2.1, so the class-level law is the finest member of the family. It was found the other way around: exhaustive computation on Q_3 and Q_4 (all $132 + 2\,459\,160$ classes, from all $42\,467\,328$ spanning trees of Q_4) revealed the pattern $N = d \cdot 2^{m-2s}$ with s equal to the cycle rank of the support of U , and the identity $m - 2\text{rank} = |S| - 1$ (an exercise with the edge budgets) collapsed it to the statement above, after which the proof was forced.

Proposition 4.2 (polarization). *Decompositions are themselves lifts: if (T, F) decomposes (S, U) , then placing $T \cap M$ on level 0 and $F \cap M$ on level 1 (with D on both) yields a spanning tree of $G \times K_2$ with data (S, U) . Hence $d(S, U)$ counts the polarized trees in their class — those whose level-0 graph is a spanning tree of G and whose level-1 graph is an S -rooted forest.*

Proof. Level 0 carries $D \cup (T \cap M) = T$, connected and acyclic; level 1 carries $D \cup (F \cap M) = F$, a forest with $|S|$ components, one per vertex of S . Each rung $u \in S$ joins the single level-0 component to the level-1 component containing u , and those components are pairwise distinct, so the component graph is a star — a tree — and the lift is a spanning tree. Conversely a tree whose level-0 graph is spanning connected and whose level-1 graph is S -rooted reads off a decomposition. $\square$

Proposition 4.3 (explicit bijection when the support is a tree). *If the support of U is a spanning tree of G (equivalently $m = |S| - 1$), then $d \leq 1$; when $d = 1$, every one of the 2^m level assignments yields a spanning tree, and the identity map on the $m = |S| - 1$ free bits, with the polarized lift as base point, realizes Theorem 4.1 bijectively.*

Proof sketch. When $\text{supp } U$ is a spanning tree, T is forced to be that tree and $F = D$, so $d = 1$ iff D is S -rooted. For validity of every assignment, induct: contract a leaf edge of the support lying in D (harmless on both sides), and otherwise a leaf edge $e = (u, \ell) \in M$ has its leaf ℓ isolated in D , hence $\ell \in S$; whichever level e is assigned, the fiber of ℓ attaches to the rest by e and the rung at ℓ , and deleting the fiber reduces to the same situation with $(S \setminus \ell, M \setminus e)$. Both choices of the bit extend every valid smaller assignment, giving the doubling and the explicit bijection. $\square$

5. OPEN PROBLEMS

All counting in the program is now proven, down to the finest stratum. The sole remaining gap between the results above and a fully bijective proof of (1) is a *map*:

Problem 5.1. *For each class (S, U) , construct an explicit bijection*

$\{\text{spanning trees of } G \times K_2 \text{ with data } (S, U)\} \longleftrightarrow \{\text{decompositions } (T, F)\} \times \{0, 1\}^{|S|-1}$, compatible with Proposition 4.2 (each tree's image should name its polarized normal form). Proposition 4.3 solves the acyclic-support classes; empirically the fibers over decompositions are (usually) cosets of a subspace of the bit space $\mathbb{F}_2^M$, so a straightening algorithm by cycle-by-cycle rerouting is the natural candidate shape. By Theorem 4.1 and Corollary 2.3, such a construction, iterated over the n directions, is a bijective proof of Stanley's formula, resolving the bijective form of [2, Example 5.6.10].

Problem 5.2. *Relate the straightening of Problem 5.1 to Bernardi's spin-flip independence [1, Thm. 1]: his inductive rerouting realizes free bits on rooted forests; ours must realize $|S| - 1$ free bits per class on unrooted trees. A dictionary between the two would likely supply the map.*

REFERENCES

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